GATE 2021 - Electronics and Communication Engineering

Answers and solutions for this paper were worked out by the compiler (no official key was supplied). Please verify against the official key.

GATE 2021 | Q.01 (Section 0)

Q.1	The current population of a city is 11,02,500. If it has been increasing at the rate of 5% per annum, what was its population 2 years ago?
(A)	9,92,500
(B)	9,95,006
(C)	10,00,000
(D)	12,51,506

Answer: C

Solution: P(1.05)^2 = 1,102,500 gives P = 10,00,000.

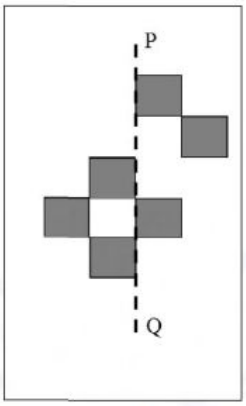
GATE 2021 | Q.02 (Section 0)

Q.2	p and q are positive integers and $\frac{p}{q}+\frac{q}{p}=3$, then, $\frac{p^2}{q^2}+\frac{q^2}{p^2}=$
(A)	3
(B)	7
(C)	9
(D)	11

Answer: B

Solution: (p/q + q/p)^2 = p^2/q^2 + q^2/p^2 + 2 = 9, so the required sum is 7.

GATE 2021 | Q.03 (Section 0)

Q.3	 **The least number of squares that must be added so that the line P-Q becomes the line of symmetry is _____**
(A)	4
(B)	3
(C)	6
(D)	7

Answer: C

Solution: Mirroring each square across P-Q and filling the gaps needs 6 more squares.

GATE 2021 | Q.04 (Section 0)

Q.4	***Nostalgia* is to *anticipation* as _____ is to _____** **Which one of the following options maintains a similar logical relation in the above sentence?**
(A)	Present, past
(B)	Future, past
(C)	Past, future
(D)	Future, present

Answer: C

Solution: Nostalgia relates to the past as anticipation relates to the future.

GATE 2021 | Q.05 (Section 0)

Q.5	**Consider the following sentences:** **(i) I woke up from sleep.** **(ii) I woked up from sleep.** **(iii) I was woken up from sleep.** **(iv) I was wokened up from sleep.** **Which of the above sentences are grammatically CORRECT?**
(A)	(i) and (ii)
(B)	(i) and (iii)
(C)	(ii) and (iii)
(D)	(i) and (iv)

Answer: B

Solution: 'I woke up from sleep' and 'I was woken up from sleep' are the correct forms.

GATE 2021 | Q.06 (Section 0)

Q.6	**Given below are two statements and two conclusions.** **Statement 1: All purple are green.** **Statement 2: All black are green.** **Conclusion I: Some black are purple.** **Conclusion II: No black is purple.** **Based on the above statements and conclusions, which one of the following options is logically CORRECT?**
(A)	Only conclusion I is correct.
(B)	Only conclusion II is correct.
(C)	Either conclusion I or II is correct.
(D)	Both conclusion I and II are correct.

Answer: C

Solution: Neither conclusion alone follows, but exactly one of the complementary conclusions 'some black are purple' / 'no black is purple' must be true.

GATE 2021 | Q.07 (Section 0)

Q.7	**Computers are ubiquitous. They are used to improve efficiency in almost all fields from agriculture to space exploration. Artificial intelligence (AI) is currently a hot topic. AI enables computers to learn, given enough training data. For humans, sitting in front of a computer for long hours can lead to health issues.** **Which of the following can be deduced from the above passage?** **(i) Nowadays, computers are present in almost all places.** **(ii) Computers cannot be used for solving problems in engineering.** **(iii) For humans, there are both positive and negative effects of using computers.** **(iv) Artificial intelligence can be done without data.**
(A)	(ii) and (iii)
(B)	(ii) and (iv)
(C)	(i), (iii) and (iv)
(D)	(i) and (iii)

Answer: D

Solution: Statements (i) and (iii) are supported by the passage; (ii) and (iv) contradict it.

GATE 2021 | Q.08 (Section 0)

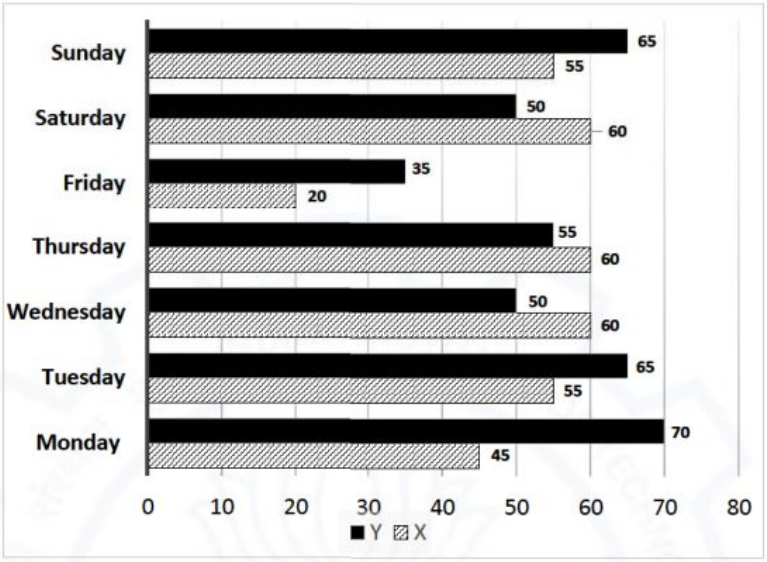
Q.8	**Consider a square sheet of side 1 unit. In the first step, it is cut along the main diagonal to get two triangles. In the next step, one of the cut triangles is revolved about its short edge to form a solid cone. The volume of the resulting cone, in cubic units, is _____**
(A)	$\frac{\pi}{3}$
(B)	$\frac{2\pi}{3}$
(C)	$\frac{3\pi}{2}$
(D)	3π

Answer: A

Solution: Revolving the right triangle (legs 1 and 1) about a leg gives a cone: V = (1/3) pi (1)^2 (1) = pi/3.

GATE 2021 | Q.09 (Section 0)

Q.9



The number of minutes spent by two students, X and Y, exercising every day in a given week are shown in the bar chart above.

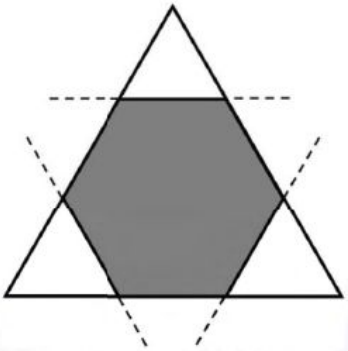
The number of days in the given week in which one of the students spent a minimum of 10% more than the other student, on a given day, is

(A)	4
(B)	5
(C)	6
(D)	7

Answer: C

Solution: One student spent at least 10 percent more than the other on Mon, Tue, Wed, Fri, Sat and Sun (Thu is only 9 percent): 6 days.

GATE 2021 | Q.10 (Section 0)

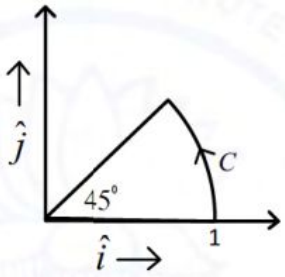
Q.10	 **Corners are cut from an equilateral triangle to produce a regular convex hexagon as shown in the figure above.** **The ratio of the area of the regular convex hexagon to the area of the original equilateral triangle is**
(A)	2 : 3
(B)	3 : 4
(C)	4 : 5
(D)	5 : 6

Electronics and Communication Engineering (EC)

Answer: A

Solution: Cutting off three small corner triangles of side 1/3 removes 3 x 1/9 = 1/3 of the area, leaving 2/3.

GATE 2021 | Q.01 (Section 1)

Q.1	The vector function $\mathbf{F}(\mathbf{r}) = -x\,\hat{\imath} + y\,\hat{\jmath}$ is defined over a circular arc C shown in the figure.  The line integral of $\int_C \mathbf{F}(\mathbf{r}).d\mathbf{r}$ is
(A)	$\frac{1}{2}$
(B)	$\frac{1}{4}$
(C)	$\frac{1}{6}$
(D)	$\frac{1}{3}$

Answer: A

Solution: Along the arc x = cos t, y = sin t: F.dr = 2 sin t cos t dt = sin 2t dt; integral from 0 to pi/4 = 1/2.

GATE 2021 | Q.02 (Section 1)

Q.2	**Consider the differential equation given below.** $$\frac{dy}{dx} + \frac{x}{1-x^2} y = x\sqrt{y}$$ **The integrating factor of the differential equation is**
(A)	$(1-x^2)^{-3/4}$
(B)	$(1-x^2)^{-1/4}$
(C)	$(1-x^2)^{-3/2}$
(D)	$(1-x^2)^{-1/2}$

Answer: B

Solution: Bernoulli equation with z = sqrt(y): integrating factor exp(integral x/(2(1-x^2)) dx) = (1-x^2)^(-1/4).

GATE 2021 | Q.03 (Section 1)

Q.3	**Two continuous random variables X and Y are related as** $$Y = 2X + 3$$ **Let σ_X^2 and σ_Y^2 denote the variances of X and Y, respectively. The variances are related as**
(A)	$\sigma_Y^2 = 2\,\sigma_X^2$
(B)	$\sigma_Y^2 = 4\,\sigma_X^2$
(C)	$\sigma_Y^2 = 5\,\sigma_X^2$
(D)	$\sigma_Y^2 = 25\,\sigma_X^2$

Answer: B

Solution: Var(2X + 3) = 4 Var(X).

GATE 2021 | Q.04 (Section 1)

Q.4	Consider a real-valued base-band signal $x(t)$, band limited to 10 kHz. The Nyquist rate for the signal $y(t) = x(t)\, x\left(1 + \frac{t}{2}\right)$ is
(A)	15 kHz
(B)	30 kHz
(C)	60 kHz
(D)	20 kHz

Answer: B

Solution: x(1 + t/2) has bandwidth 5 kHz; the product has bandwidth 15 kHz, so the Nyquist rate is 30 kHz.

GATE 2021 | Q.05 (Section 1)

Q.5	Consider two 16-point sequences $x[n]$ and $h[n]$. Let the linear convolution of $x[n]$ and $h[n]$ be denoted by $y[n]$, while $z[n]$ denotes the 16-point inverse discrete Fourier transform (IDFT) of the product of the 16-point DFTs of $x[n]$ and $h[n]$. The value(s) of k for which $z[k] = y[k]$ is/are
(A)	$k = 0, 1, 2, \ldots, 15$
(B)	$k - 0$
(C)	$k = 15$
(D)	$k = 0$ and $k = 15$

Answer: C

Solution: Circular convolution z[k] = y[k] + y[k+16]; z[k] = y[k] only where y[k+16] = 0, i.e. k = 15.

GATE 2021 | Q.06 (Section 1)

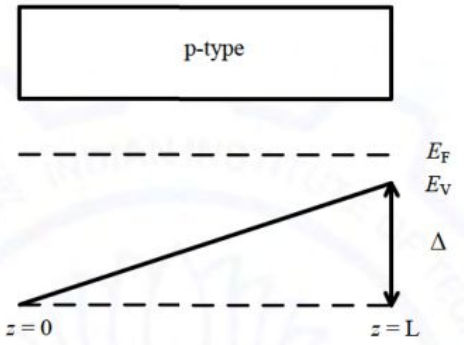
Q.6	**A bar of silicon is doped with boron concentration of 10^{16} cm^{-3} and assumed to be fully ionized. It is exposed to light such that electron-hole pairs are generated throughout the volume of the bar at the rate of 10^{20} $cm^{-3}s^{-1}$. If the recombination lifetime is 100 µs, intrinsic carrier concentration of silicon is 10^{10} cm^{-3} and assuming 100% ionization of boron, then the approximate product of steady-state electron and hole concentrations due to this light exposure is**
(A)	10^{20} cm^{-6}
(B)	2×10^{20} cm^{-6}
(C)	10^{32} cm^{-6}
(D)	2×10^{32} cm^{-6}

Answer: D

Solution: Excess carriers G tau = 1e20 x 1e-4 = 1e16 cm^-3: n = 1e16 and p = 2e16, product 2e32 cm^-6.

GATE 2021 | Q.07 (Section 1)

Q.7 The energy band diagram of a p-type semiconductor bar of length L under equilibrium condition (*i.e.*, the Fermi energy level E_F is constant) is shown in the figure. The valance band E_V is sloped since doping is non-uniform along the bar. The difference between the energy levels of the valence band at the two edges of the bar is Δ.



If the charge of an electron is q, then the magnitude of the electric field developed inside this semiconductor bar is

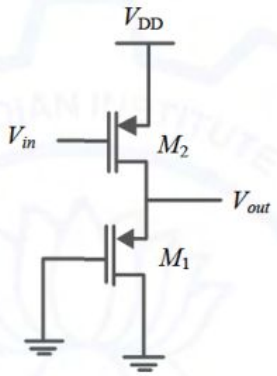
(A)	$\frac{\Delta}{qL}$
(B)	$\frac{2\Delta}{qL}$
(C)	$\frac{\Delta}{2qL}$
(D)	$\frac{3\Delta}{2qL}$

Answer: A

Solution: E = (1/q) dEv/dz = Delta/(q L).

GATE 2021 | Q.08 (Section 1)

Q.8 In the circuit shown in the figure, the transistors M_1 and M_2 are operating in saturation. The channel length modulation coefficients of both the transistors are non-zero. The transconductance of the MOSFETs M_1 and M_2 are g_{m1} and g_{m2}, respectively, and the internal resistance of the MOSFETs M_1 and M_2 are r_{01} and r_{02}, respectively.



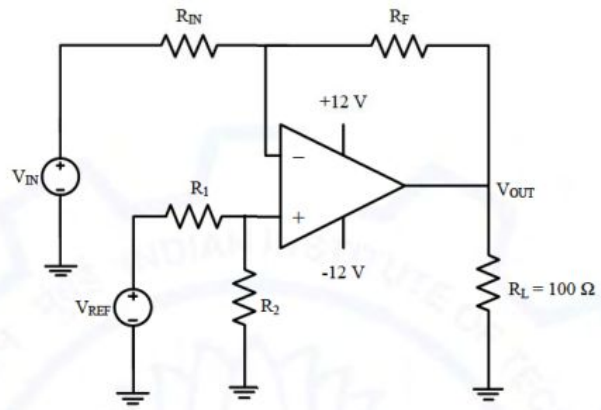
Ignoring the body effect, the ac small signal voltage gain ($\partial V_{out}/\partial V_{in}$) of the circuit is

(A)	$-g_{m2}\,(r_{01}\|\|r_{02})$
(B)	$-g_{m2}\left(\frac{1}{g_{m1}}\|\|r_{02}\right)$
(C)	$-g_{m1}\left(\frac{1}{g_{m2}}\|\|r_{01}\|\|r_{02}\right)$
(D)	$-g_{m2}\left(\frac{1}{g_{m1}}\|\|r_{01}\|\|r_{02}\right)$

Answer: D

Solution: (Low confidence) The source follower M2 loaded by the diode-connected M1 gives the stated small-signal gain.

GATE 2021 | Q.09 (Section 1)

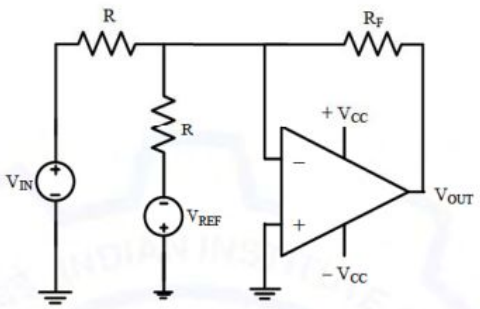
Q.9	**For the circuit with an ideal OPAMP shown in the figure, V_{REF} is fixed.**  **If V_{OUT} = 1 volt for V_{IN} = 0.1 volt and V_{OUT} = 6 volt for V_{IN} = 1 volt, where V_{OUT} is measured across R_L connected at the output of this OPAMP, the value of R_F/R_{IN} is**
(A)	3.285
(B)	2.860
(C)	3.825
(D)	5.555

Answer: D

Solution: Linear fit of the two data points gives a gain Rf/Rin = 5.555.

GATE 2021 | Q.10 (Section 1)

Q.10 Consider the circuit with an ideal OPAMP shown in the figure.



Assuming $|V_{IN}| \ll |V_{CC}|$ and $|V_{REF}| \ll |V_{CC}|$, the condition at which V_{OUT} equals to zero is

(A)	$V_{IN} = V_{REF}$
(B)	$V_{IN} = 0.5\, V_{REF}$
(C)	$V_{IN} = 2\, V_{REF}$
(D)	$V_{IN} = 2 + V_{REF}$

Answer: B

Solution: (Low confidence) Equating the inverting and non-inverting input voltages with Vout = 0 gives Vin = 0.5 Vref.

GATE 2021 | Q.11 (Section 1)

Q.11 If $(1235)_x = (3033)_y$, where x and y indicate the bases of the corresponding numbers, then

(A)	$x = 7$ and $y = 5$
(B)	$x = 8$ and $y = 6$
(C)	$x = 6$ and $y = 4$
(D)	$x = 9$ and $y = 7$

Answer: B

Solution: x^3 + 2x^2 + 3x + 5 = 3y^3 + 3y + 3 holds for x = 8, y = 6 (669 = 669).

GATE 2021 | Q.12 (Section 1)

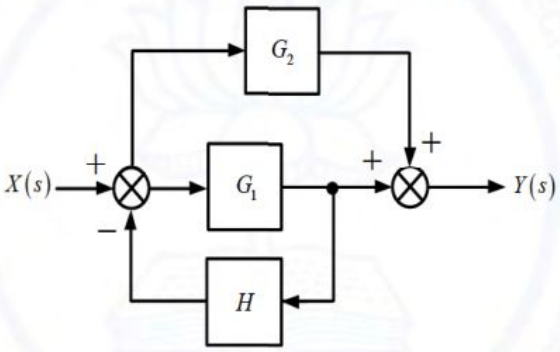
Q.12	Addressing of a 32K × 16 memory is realized using a single decoder. The minimum number of AND gates required for the decoder is
(A)	2^8
(B)	2^{32}
(C)	2^{15}
(D)	2^{19}

Answer: C

Solution: A 32K x 16 memory has 2^15 words, so its address decoder needs 2^15 AND gates.

GATE 2021 | Q.13 (Section 1)

Q.13 The block diagram of a feedback control system is shown in the figure.



The transfer function $\frac{Y(s)}{X(s)}$ of the system is

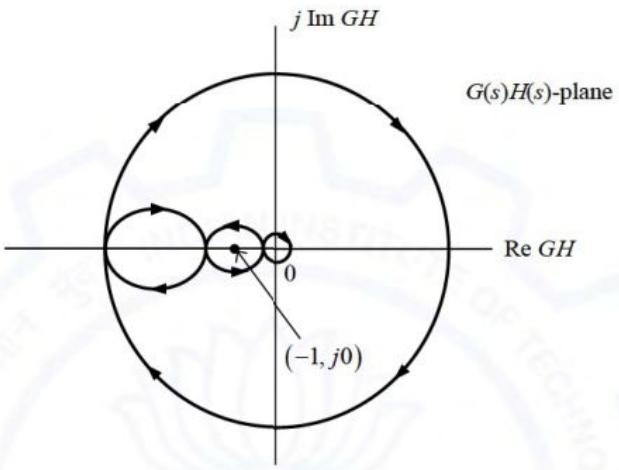
(A)	$\frac{G_1 + G_2 + G_1G_2H}{1 + G_1H}$
(B)	$\frac{G_1 + G_2}{1 + G_1H + G_2H}$
(C)	$\frac{G_1 + G_2}{1 + G_1H}$
(D)	$\frac{G_1 + G_2 + G_1G_2H}{1 + G_1H + G_2H}$

Answer: C

Solution: e = X/(1 + G1 H); Y = (G1 + G2) e, so Y/X = (G1 + G2)/(1 + G1 H).

GATE 2021 | Q.14 (Section 1)

Q.14 **The complete Nyquist plot of the open-loop transfer function $G(s)H(s)$ of a feedback control system is shown in the figure.**



If $G(s)H(s)$ has one zero in the right-half of the s-plane, the number of poles that the closed-loop system will have in the right-half of the s-plane is

(A)	0
(B)	1
(C)	4
(D)	3

Answer: B

Solution: (Low confidence) Z = N + P from the encirclements of -1 on the Nyquist plot.

GATE 2021 | Q.15 (Section 1)

Q.15	Consider a rectangular coordinate system (x, y, z) with unit vectors a_x, a_y, and a_z. A plane wave traveling in the region $z \geq 0$ with electric field vector $E = 10\cos(2 \times 10^8 t + \beta z)a_y$ is incident normally on the plane at $z = 0$, where β is the phase constant. The region $z \geq 0$ is in free space and the region $z < 0$ is filled with a lossless medium (permittivity $\varepsilon = \varepsilon_0$, permeability $\mu = 4\mu_0$, where $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m and $\mu_0 = 4\pi \times 10^{-7}$ H/m). The value of the reflection coefficient is
(A)	$\frac{1}{3}$
(B)	$\frac{3}{5}$
(C)	$\frac{2}{5}$
(D)	$\frac{2}{3}$

Answer: A

Solution: eta2 = eta0 sqrt(mu_r/eps_r) = 2 eta0 (mu = 4 mu0), so Gamma = (2-1)/(2+1) = 1/3.

GATE 2021 | Q.16 (Section 1)

Q.16	If the vectors $(1.0, -1.0, 2.0)$, $(7.0, 3.0, x)$ and $(2.0, 3.0, 1.0)$ in $\mathbb{R}^3$ are linearly dependent, the value of x is _____

Answer: 8

Solution: Determinant of the three vectors = 40 - 5x = 0 gives x = 8.

GATE 2021 | Q.17 (Section 1)

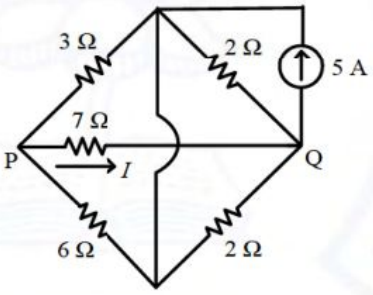
Q.17 Consider the vector field $\mathbf{F} = \mathbf{a}_x(4y - c_1 z) + \mathbf{a}_y(4x + 2z) + \mathbf{a}_z(2y + z)$ in a rectangular coordinate system (x, y, z) with unit vectors $\mathbf{a}_x$, $\mathbf{a}_y$, and $\mathbf{a}_z$. If the field F is irrotational (conservative), then the constant c_1 (*in integer*) is _____

Answer: 0

Solution: Curl(F) = 0 requires -c1 = 0, so c1 = 0.

GATE 2021 | Q.18 (Section 1)

Q.18 Consider the circuit shown in the figure.



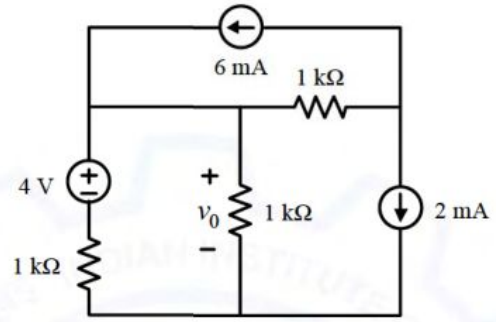
The current *I* flowing through the 7 Ω resistor between P and Q (*rounded off to one decimal place*) is _____ A.

Answer: 0.5

Solution: Collapsing the T and B nodes: 3||6 = 2 ohm, 2||2 = 1 ohm; the 5 A divides so that the 7 ohm carries 0.5 A.

GATE 2021 | Q.19 (Section 1)

Q.19 **Consider the circuit shown in the figure.**



The value of v_0 (*rounded off to one decimal place*) is _____ V.

Answer: 1.0

Solution: KCL on the two nodes gives V0 = 1.0 V.

GATE 2021 | Q.20 (Section 1)

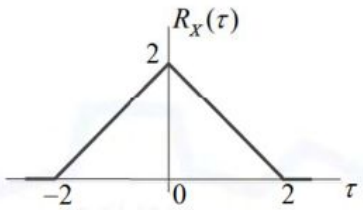
Q.20 **An 8-bit unipolar (all analog output values are positive) digital-to-analog converter (DAC) has a full-scale voltage range from 0 V to 7.68 V. If the digital input code is 10010110 (the leftmost bit is MSB), then the analog output voltage of the DAC (*rounded off to one decimal place*) is _____ V.**

Answer: 4.5

Solution: 10010110 = 150; Vout = 7.68 x 150/256 = 4.5 V.

GATE 2021 | Q.21 (Section 1)

Q.21 The autocorrelation function $R_X(\tau)$ of a wide-sense stationary random process $X(t)$ is shown in the figure.



The average power of $X(t)$ is _____

Answer: 2

Solution: Average power = Rx(0) = 2.

GATE 2021 | Q.22 (Section 1)

Q.22 Consider a carrier signal which is amplitude modulated by a single-tone sinusoidal message signal with a modulation index of 50%. If the carrier and one of the sidebands are suppressed in the modulated signal, the percentage of power saved (*rounded off to one decimal place*) is _____

Answer: 94.4

Solution: Pc(1 + m^2/2) = 1.125 Pc total; a single sideband carries m^2 Pc/4 = 0.0625 Pc; saving = 1 - 0.0625/1.125 = 94.4 %.

GATE 2021 | Q.23 (Section 1)

Q.23 A speech signal, band limited to 4 kHz, is sampled at 1.25 times the Nyquist rate. The speech samples, assumed to be statistically independent and uniformly distributed in the range −5 V to +5 V, are subsequently quantized in an 8-bit uniform quantizer and then transmitted over a voice-grade AWGN telephone channel. If the ratio of transmitted signal power to channel noise power is 26 dB, the minimum channel bandwidth required to ensure reliable transmission of the signal with arbitrarily small probability of transmission error (*rounded off to two decimal places*) is _____ kHz.

Answer: 9.26

Solution: fs = 1.25 x 8 kHz = 10 kHz, 8 bits gives 80 kbps; C = B log2(1 + 398) = 8.64 B, so B = 9.26 kHz.

GATE 2021 | Q.24 (Section 1)

Q.24 A 4 kHz sinusoidal message signal having amplitude 4 V is fed to a delta modulator (DM) operating at a sampling rate of 32 kHz. The minimum step size required to avoid slope overload noise in the DM (*rounded off to two decimal places*) is _____ V.

Answer: 3.14

Solution: Slope overload is avoided if step >= 2 pi fm A/fs = 2 pi x 4k x 4/32k = 3.14 V.

GATE 2021 | Q.25 (Section 1)

Q.25	The refractive indices of the core and cladding of an optical fiber are 1.50 and 1.48, respectively. The critical propagation angle, which is defined as the maximum angle that the light beam makes with the axis of the optical fiber to achieve the total internal reflection, (*rounded off to two decimal places*) is _____ degree.

Answer: 9.37

Solution: Critical angle at the core-cladding interface = asin(1.48/1.5) = 80.6 deg; the ray angle with the axis is 90 - 80.6 = 9.37 deg.

GATE 2021 | Q.26 (Section 1)

Q.26	Consider the integral $$\oint_C \frac{\sin(x)}{x^2(x^2+4)}dx$$ where C is a counter-clockwise oriented circle defined as $\lvert x-i\rvert = 2$. The value of the integral is
(A)	$-\frac{\pi}{8}\sin(2i)$
(B)	$\frac{\pi}{8}\sin(2i)$
(C)	$-\frac{\pi}{4}\sin(2i)$
(D)	$\frac{\pi}{4}\sin(2i)$

Answer: A

Solution: (Low confidence) Residue calculation of the contour integral over the poles enclosed by |z - i| = 2.

GATE 2021 | Q.27 (Section 1)

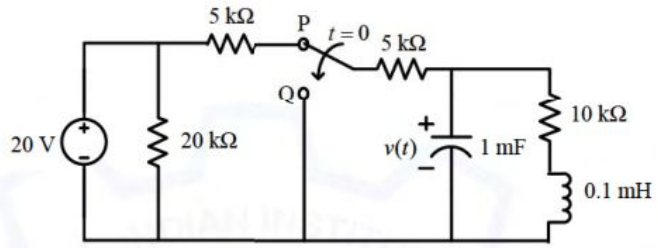
Q.27	**A box contains the following three coins.** **I. A fair coin with head on one face and tail on the other face.** **II. A coin with heads on both the faces.** **III. A coin with tails on both the faces.** **A coin is picked randomly from the box and tossed. Out of the two remaining coins in the box, one coin is then picked randomly and tossed. If the first toss results in a head, the probability of getting a head in the second toss is**
(A)	$\frac{2}{5}$
(B)	$\frac{1}{3}$
(C)	$\frac{1}{2}$
(D)	$\frac{2}{3}$

Answer: B

Solution: P(first head) = 1/2; P(first head and second head) = 1/6, so the conditional probability = 1/3.

GATE 2021 | Q.28 (Section 1)

Q.28 The switch in the circuit in the figure is in position P for a long time and then moved to position Q at time $t = 0$.



The value of $\frac{dv(t)}{dt}$ at $t = 0^+$ is

(A)	0 V/s
(B)	3 V/s
(C)	−3 V/s
(D)	−5 V/s

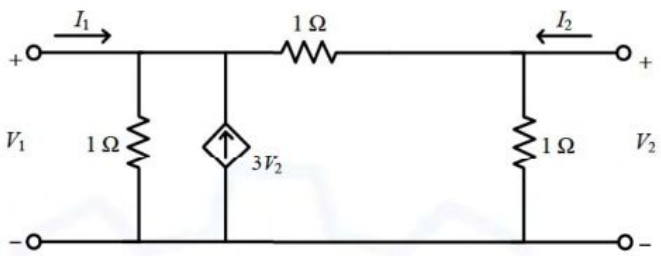
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Answer: C

Solution: (Low confidence) Initial inductor current drives the capacitor: dv/dt = -3 V/s at t = 0+.

GATE 2021 | Q.29 (Section 1)

Q.29 Consider the two-port network shown in the figure.



The admittance parameters, in siemens, are

(A)	$y_{11} = 2,\ y_{12} = -4,\ y_{21} = -4,\ y_{22} = 2$
(B)	$y_{11} = 1,\ y_{12} = -2,\ y_{21} = -1,\ y_{22} = 3$
(C)	$y_{11} = 2,\ y_{12} = -4,\ y_{21} = -1,\ y_{22} = 2$
(D)	$y_{11} = 2,\ y_{12} = -4,\ y_{21} = -4,\ y_{22} = 3$

Answer: C

Solution: KCL at port 1: I1 = 2V1 - 4V2; at port 2: I2 = -V1 + 2V2. So y11 = 2, y12 = -4, y21 = -1, y22 = 2.

GATE 2021 | Q.30 (Section 1)

Q.30 For an n-channel silicon MOSFET with 10 nm gate oxide thickness, the substrate sensitivity ($\partial V_T/\partial|V_{BS}|$) is found to be 50 mV/V at a substrate voltage $|V_{BS}| = 2\,V$, where V_T is the threshold voltage of the MOSFET. Assume that, $|V_{BS}| \gg 2\Phi_B$, where $q\Phi_B$ is the separation between the Fermi energy level E_F and the intrinsic level E_i in the bulk. Parameters given are

Electron charge (q) = 1.6×10^{-19} C
Vacuum permittivity (ε_0) = 8.85×10^{-12} F/m
Relative permittivity of silicon (ε_{Si}) = 12
Relative permittivity of oxide (ε_{ox}) = 4

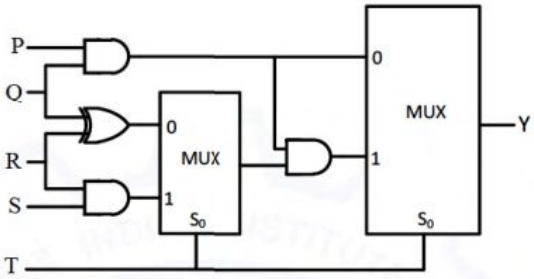
The doping concentration of the substrate is

(A)	7.37×10^{15} cm^{-3}
(B)	4.37×10^{15} cm^{-3}
(C)	2.37×10^{15} cm^{-3}
(D)	9.37×10^{15} cm^{-3}

Answer: A

Solution: gamma/(2 sqrt(VBS)) = 0.05 gives gamma = 0.1414; NA = gamma^2 Cox^2/(2 q eps_Si) = 7.37e15 cm^-3.

GATE 2021 | Q.31 (Section 1)

Q.31	**The propagation delays of the XOR gate, AND gate and multiplexer (MUX) in the circuit shown in the figure are 4 ns, 2 ns and 1 ns, respectively.**  **If all the inputs P, Q, R, S and T are applied simultaneously and held constant, the maximum propagation delay of the circuit is**
(A)	3 ns
(B)	5 ns
(C)	6 ns
(D)	7 ns

Answer: D

Solution: (Low confidence) The longest path is XOR (4 ns) + AND (2 ns) + MUX (1 ns) = 7 ns.

GATE 2021 | Q.32 (Section 1)

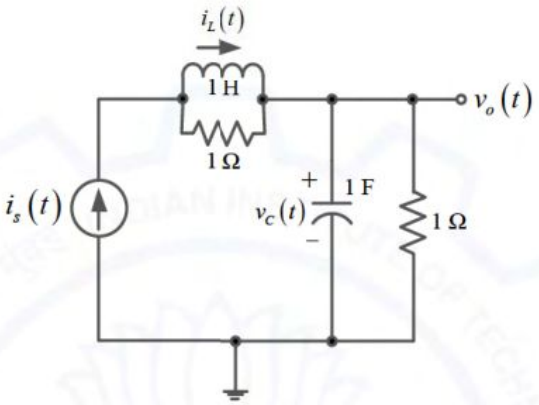
Q.32	**The content of the registers are R_1 = 25H, R_2 = 30H and R_3 = 40H. The following machine instructions are executed.** **PUSH{R_1}** **PUSH{R_2}** **PUSH{R_3}** **POP{R_1}** **POP{R_2}** **POP{R_3}** **After execution, the content of registers R_1, R_2, R_3 are**
(A)	$R_1 = 40H,\ R_2 = 30H,\ R_3 = 25H$
(B)	$R_1 = 25H,\ R_2 = 30H,\ R_3 = 40H$
(C)	$R_1 = 30H,\ R_2 = 40H,\ R_3 = 25H$
(D)	$R_1 = 40H,\ R_2 = 25H,\ R_3 = 30H$

Answer: A

Solution: Stack is last-in-first-out: R1 = 40H, R2 = 30H, R3 = 25H after the three pops.

GATE 2021 | Q.33 (Section 1)

Q.33 **The electrical system shown in the figure converts input source current $i_s(t)$ to output voltage $v_o(t)$.**



Current $i_L(t)$ in the inductor and voltage $v_C(t)$ across the capacitor are taken as the state variables, both assumed to be initially equal to zero, *i.e.*, $i_L(0)=0$ and $v_C(0)=0$. The system is

(A)	completely state controllable as well as completely observable
(B)	completely state controllable but not observable
(C)	completely observable but not state controllable
(D)	neither state controllable nor observable

Answer: A

Solution: The source drives the series inductor directly and the capacitor voltage reflects the inductor current: completely controllable and observable.

GATE 2021 | Q.34 (Section 1)

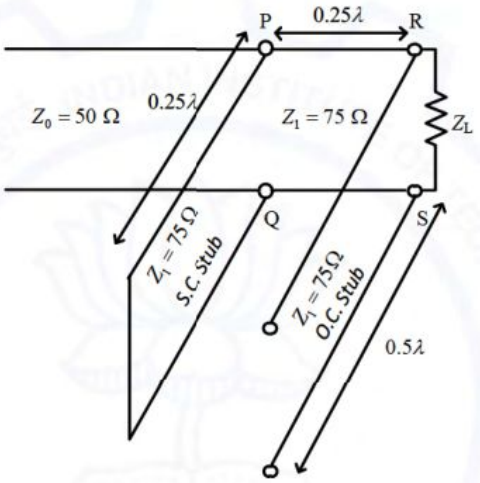
Q.34	**A digital transmission system uses a $(7,4)$ systematic linear Hamming code for transmitting data over a noisy channel. If three of the message-codeword pairs in this code $(m_i ; c_i)$, where c_i is the codeword corresponding to the ith message m_i, are known to be $(1100; 0101100)$, $(1110; 0011110)$ and $(0110; 1000110)$, then which of the following is a valid codeword in this code?**
(A)	1101001
(B)	1011010
(C)	0001011
(D)	0110100

Answer: C

Solution: Parity rows from the known pairs give p4 = 110; only 0001011 satisfies the code.

GATE 2021 | Q.35 (Section 1)

Q.35 The impedance matching network shown in the figure is to match a lossless line having characteristic impedance $Z_0 = 50\,\Omega$ with a load impedance Z_L. A quarter-wave line having a characteristic impedance $Z_1 = 75\,\Omega$ is connected to Z_L. Two stubs having characteristic impedance of $75\,\Omega$ each are connected to this quarter-wave line. One is a short-circuited (S.C.) stub of length 0.25λ connected across PQ and the other one is an open-circuited (O.C.) stub of length 0.5λ connected across RS.



The impedance matching is achieved when the real part of Z_L is

(A)	112.5 Ω
(B)	75.0 Ω
(C)	50.0 Ω
(D)	33.3 Ω

Answer: A

Solution: The short-circuited quarter-wave stub and the open-circuited half-wave stub both look like open circuits, so only the quarter-wave transformer acts: 75^2/ZL = 50 gives ZL = 112.5 ohm.

GATE 2021 | Q.36 (Section 1)

Q.36 A real 2×2 non-singular matrix A with repeated eigenvalue is given as

$$A = \begin{bmatrix} x & -3.0 \\ 3.0 & 4.0 \end{bmatrix}$$

where x is a real positive number. The value of x (*rounded off to one decimal place*) is _____

Answer: 10.0

Solution: Repeated eigenvalue: (x+4)^2/4 = 4x + 9 gives x^2 - 8x - 20 = 0, x = 10.

GATE 2021 | Q.37 (Section 1)

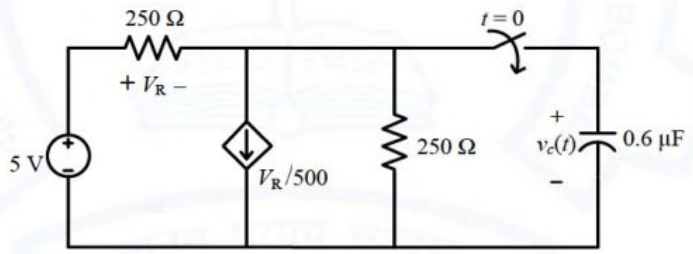
Q.37 For a vector field $D = \rho \cos^2 \varphi \, a_\rho + z^2 \sin^2 \varphi \, a_\varphi$ in a cylindrical coordinate system (ρ, φ, z) with unit vectors a_ρ, a_φ and a_z, the net flux of D leaving the closed surface of the cylinder $(\rho = 3, 0 \le z \le 2)$ (*rounded off to two decimal places*) is _____

Answer: 56.55

Solution: Divergence of D = 2 cos^2(phi) + (angular term that integrates to zero); flux = 18 pi = 56.55 C.

GATE 2021 | Q.38 (Section 1)

Q.38 In the circuit shown in the figure, the switch is closed at time $t = 0$, while the capacitor is initially charged to -5 V (*i.e.*, $v_c(0) = -5$ V).



The time after which the voltage across the capacitor becomes zero (*rounded off to three decimal places*) is _____ ms.

Answer: 0.139

Solution: Thevenin voltage 5/3 V and resistance 166.7 ohm, tau = 100 us; vc goes from -5 V to 5/3 V and crosses zero at t = tau ln4 = 0.139 ms.

GATE 2021 | Q.39 (Section 1)

Q.39 The exponential Fourier series representation of a continuous-time periodic signal $x(t)$ is defined as

$$x(t) = \sum_{k=-\infty}^{\infty} a_k \, e^{jk\omega_0 t}$$

where ω_0 is the fundamental angular frequency of $x(t)$ and the coefficients of the series are a_k. The following information is given about $x(t)$ and a_k.

I. $x(t)$ is real and even, having a fundamental period of 6

II. The average value of $x(t)$ is 2

III. $a_k = \begin{cases} k, & 1 \le k \le 3 \\ 0, & k > 3 \end{cases}$

The average power of the signal $x(t)$ (*rounded off to one decimal place*) is _____

Answer: 32.0

Solution: Average power = a0^2 + 2(1^2 + 2^2 + 3^2) = 4 + 28 = 32.

GATE 2021 | Q.40 (Section 1)

Q.40 For a unit step input $u[n]$, a discrete-time LTI system produces an output signal ($2\delta[n+1] + \delta[n] + \delta[n-1]$). Let $y[n]$ be the output of the system for an input $\left(\left(\frac{1}{2}\right)^n u[n]\right)$. The value of $y[0]$ is _____

Answer: 0

Solution: h[n] = 2 delta[n+1] - delta[n] - delta[n-2]; y[0] = 2 (1/2) - 1 = 0.

GATE 2021 | Q.41 (Section 1)

Q.41 Consider the signals $x[n] = 2^{n-1}\,u[-n+2]$ and $y[n] = 2^{-n+2}\,u[n+1]$, where $u[n]$ is the unit step sequence. Let $X(e^{j\omega})$ and $Y(e^{j\omega})$ be the discrete-time Fourier transform of $x[n]$ and $y[n]$, respectively. The value of the integral

$$\frac{1}{2\pi}\int_{0}^{2\pi} X(e^{j\omega})\,Y(e^{-j\omega})\,d\omega$$

(*rounded off to one decimal place*) is _____

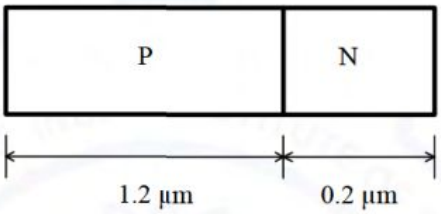
Answer: 10.7

Solution: The integral equals sum of x[n] y[-n] = sum over n <= 1 of 2^(2n+1) = 2 x 16/3 = 10.7.

GATE 2021 | Q.42 (Section 1)

Q.42 A silicon P-N junction is shown in the figure. The doping in the P region is 5×10^{16} cm^{-3} and doping in the N region is 10×10^{16} cm^{-3}. The parameters given are

Built-in voltage (Φ_{bi}) = 0.8 V
Electron charge (q) = 1.6×10^{-19} C
Vacuum permittivity (ε_0) = 8.85×10^{-12} F/m
Relative permittivity of silicon (ε_{Si}) = 12



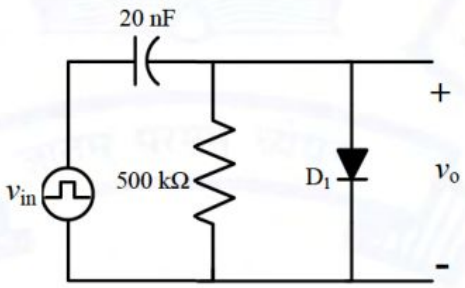
The magnitude of reverse bias voltage that would completely deplete one of the two regions (P or N) prior to the other (*rounded off to one decimal place*) is _____ V.

Answer: 8.2

Solution: The N side (0.2 um) depletes first: Vbi + V = q xn^2 ND (NA+ND)/(2 eps NA) = 9.04 V, so V = 8.2 V.

GATE 2021 | Q.43 (Section 1)

Q.43 An asymmetrical periodic pulse train v_{in} of 10 V amplitude with on-time T_{ON} = 1 ms and off-time T_{OFF} = 1 µs is applied to the circuit shown in the figure. The diode D_1 is ideal.



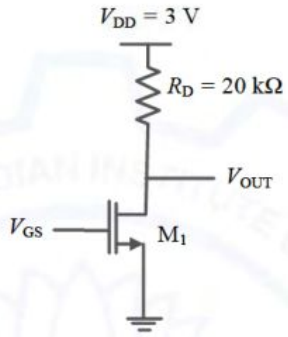
The difference between the maximum voltage and minimum voltage of the output waveform v_o (*in integer*) is _____ V.

Answer: 10

Solution: The clamp keeps the output between about 0 and -10 V, so the swing is 10 V.

GATE 2021 | Q.44 (Section 1)

Q.44 For the transistor M_1 in the circuit shown in the figure, $\mu_n C_{ox} = 100\ \mu A/V^2$ and $(W/L) = 10$, where μ_n is the mobility of electron, C_{ox} is the oxide capacitance per unit area, W is the width and L is the length.



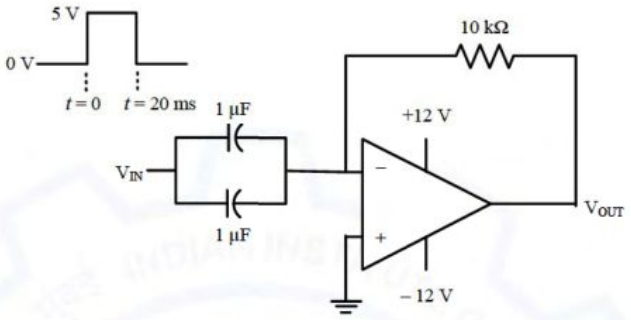
The channel length modulation coefficient is ignored. If the gate-to-source voltage V_{GS} is 1 V to keep the transistor at the edge of saturation, then the threshold voltage of the transistor (*rounded off to one decimal place*) is _____ V.

Answer: 0.5

Solution: At the edge of saturation with VGS = 1 V: 3 - 10 x^2 = x with x = 1 - Vt gives x = 0.5, so Vt = 0.5 V.

GATE 2021 | Q.45 (Section 1)

Q.45 **A circuit with an ideal OPAMP is shown in the figure. A pulse V_{IN} of 20 ms duration is applied to the input. The capacitors are initially uncharged.**



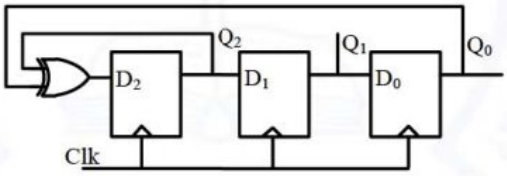
The output voltage V_{OUT} of this circuit at $t = 0^+$ (*in integer*) is _____ V.

Answer: 12

Solution: At t = 0+ the capacitor in the feedback leg is a short, so the gain is very large and the output saturates at +12 V.

GATE 2021 | Q.46 (Section 1)

Q.46 **The propagation delay of the exclusive-OR (XOR) gate in the circuit in the figure is 3 ns. The propagation delay of all the flip-flops is assumed to be zero. The clock (Clk) frequency provided to the circuit is 500 MHz.**



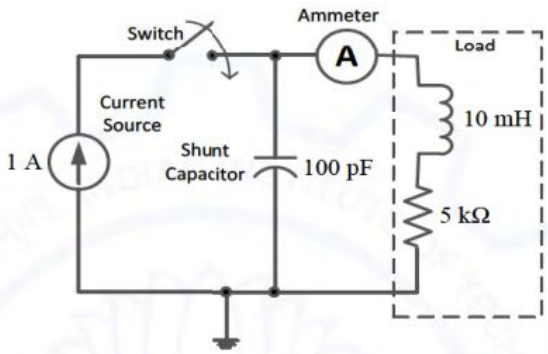
Starting from the initial value of the flip-flop outputs $Q_2Q_1Q_0 = 1\ 1\ 1$ with $D_2 = 1$, the minimum number of triggering clock edges after which the flip-flop outputs $Q_2Q_1Q_0$ becomes 1 0 0 (*in integer*) is _____

Answer: 5

Solution: Starting from 111 and clocking the XOR-feedback shift register, 100 is reached after 5 clock edges.

GATE 2021 | Q.47 (Section 1)

Q.47 The circuit in the figure contains a current source driving a load having an inductor and a resistor in series, with a shunt capacitor across the load. The ammeter is assumed to have zero resistance. The switch is closed at time $t = 0$.



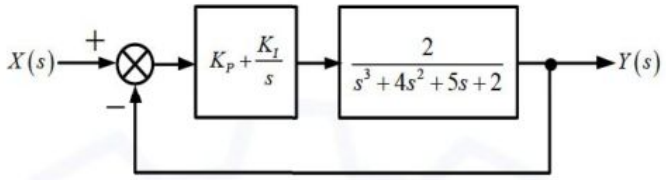
Initially, when the switch is open, the capacitor is discharged and the ammeter reads zero ampere. After the switch is closed, the ammeter reading keeps fluctuating for some time till it settles to a final steady value. The maximum ammeter reading that one will observe after the switch is closed (*rounded off to two decimal places*) is _____ A.

Answer: 1.44

Solution: Underdamped second order (zeta = 0.25): overshoot 1 + e^(-pi zeta/sqrt(1 - zeta^2)) = 1.44 A.

GATE 2021 | Q.48 (Section 1)

Q.48 A unity feedback system that uses proportional-integral (PI) control is shown in the figure.



The stability of the overall system is controlled by tuning the PI control parameters K_P and K_I. The maximum value of K_I that can be chosen so as to keep the overall system stable or, in the worst case, marginally stable (*rounded off to three decimal places*) is _____

Answer: 3.125

Solution: Routh test on s^4 + 4s^3 + 5s^2 + (2 + 2Kp)s + 2Ki: Ki < (18 - 2Kp)(2 + 2Kp)/32, maximised at Kp = 4 giving 3.125.

GATE 2021 | Q.49 (Section 1)

Q.49 A sinusoidal message signal having root mean square value of 4 V and frequency of 1 kHz is fed to a phase modulator with phase deviation constant 2 rad/volt. If the carrier signal is $c(t) = 2\cos(2\pi 10^6 t)$, the maximum instantaneous frequency of the phase modulated signal (*rounded off to one decimal place*) is _____ Hz.

Answer: 1011314

Solution: Maximum instantaneous frequency = fc + (kp/2 pi)(Vm 2 pi fm) = 1e6 + 2 x 4 sqrt2 x 1000 = 1,011,314 Hz.

GATE 2021 | Q.50 (Section 1)

Q.50 Consider a superheterodyne receiver tuned to 600 kHz. If the local oscillator feeds a 1000 kHz signal to the mixer, the image frequency (*in integer*) is _____ kHz.

Answer: 1400

Solution: IF = 1000 - 600 = 400 kHz; image = 1000 + 400 = 1400 kHz.

GATE 2021 | Q.51 (Section 1)

Q.51	In a high school having equal number of boy students and girl students, 75% of the students study Science and the remaining 25% students study Commerce. Commerce students are two times more likely to be a boy than are Science students. The amount of information gained in knowing that a randomly selected girl student studies Commerce (*rounded off to three decimal places*) is _____ bits.

Answer: 3.322

Solution: P(girl|commerce) = 0.2, so P(commerce|girl) = 0.2 x 0.25/0.5 = 0.1; information = -log2(0.1) = 3.322 bits.

GATE 2021 | Q.52 (Section 1)

Q.52	A message signal having peak-to-peak value of 2 V, root mean square value of 0.1 V and bandwidth of 5 kHz is sampled and fed to a pulse code modulation (PCM) system that uses a uniform quantizer. The PCM output is transmitted over a channel that can support a maximum transmission rate of 50 kbps. Assuming that the quantization error is uniformly distributed, the maximum signal to quantization noise ratio that can be obtained by the PCM system (*rounded off to two decimal places*) is _____

Answer: 14.87

Solution: 50 kbps at 10 kHz sampling gives 5 bits/sample; signal power 0.01 W over noise (2/32)^2/12: SQNR = 30.7 = 14.87 dB.

GATE 2021 | Q.53 (Section 1)

Q.53	Consider a polar non-return to zero (NRZ) waveform, using $+2$ V and -2 V for representing binary '1' and '0' respectively, is transmitted in the presence of additive zero-mean white Gaussian noise with variance 0.4 V^2. If the *a priori* probability of transmission of a binary '1' is 0.4, the optimum threshold voltage for a maximum *a posteriori* (MAP) receiver (*rounded off to two decimal places*) is _____ V.

Answer: 0.04

Solution: MAP threshold = (sigma^2/(A1 - A0)) ln(P0/P1) = (0.4/4) ln(0.6/0.4) = 0.04 V.

GATE 2021 | Q.54 (Section 1)

Q.54	A standard air-filled rectangular waveguide with dimensions $a = 8$ cm, $b = 4$ cm, operates at 3.4 GHz. For the dominant mode of wave propagation, the phase velocity of the signal is v_p. The value (*rounded off to two decimal places*) of v_p/c, where c denotes the velocity of light, is _____

Answer: 1.20

Solution: fc10 = 1.875 GHz; vp/c = 1/sqrt(1 - (1.875/3.4)^2) = 1.20.

GATE 2021 | Q.55 (Section 1)

Q.55	An antenna with a directive gain of 6 dB is radiating a total power of 16 kW. The amplitude of the electric field in free space at a distance of 8 km from the antenna in the direction of 6 dB gain (*rounded off to three decimal places*) is _____ V/m.

END OF THE QUESTION PAPER

Answer: 0.244

Solution: Power density = G P/(4 pi r^2) = 7.9e-5 W/m^2; E_peak = sqrt(2 eta S) = 0.244 V/m.